Phuc Nguyen Tran Thien

AI & Machine Learning · Computer Vision · Research & Engineering

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Portfolio / GitHub / LinkedIn / Google Scholar / ORCID

Research Assistant with AI engineering experience in computer vision, multimodal learning, OCR and semantic retrieval. Experience includes training and evaluation pipelines, inference software and application APIs. M.Sc. student and co-author of six publications. Open to AI/ML engineering roles, research assistantships and international graduate opportunities.

TECHNICAL SKILLS

AI/ML: Python, PyTorch, scikit-learn, OpenCV, MediaPipe, ONNX Runtime; model training, dataset audits, error analysis and reproducible evaluation.
Engineering: FastAPI, REST APIs, WebSocket, Docker, Git, Linux, Flutter.
Retrieval: OCR, Sentence Transformers, Elasticsearch, MongoDB, BM25 and dense retrieval.

EXPERIENCE

Research Assistant · Ho Chi Minh City Open University Aug 2025-Present

• Develop computer vision and multimodal models for visual speech, Vietnamese isolated signs and smartphone chemical sensing.

• Build PyTorch experiments, inspect dataset splits and model errors, and develop reusable inference tools and application prototypes.

• Co-author six publications, including two first-author IEEE JCSSE papers and an article in *Analytica Chimica Acta*.

AI Engineer Intern · FPT Software Sep-Dec 2024

• Built OCR and semantic-search components using EasyOCR, Sentence Transformers, Elasticsearch and MongoDB.

• Integrated hybrid BM25/dense retrieval, metadata filtering and REST APIs with application teams.

SELECTED AI PROJECTS

AI-Chemistry - Developed colorimetric-strip preprocessing, multi-task concentration prediction and a FastAPI inference service. Connects model experiments with an application-facing API.

LipTrace - Built dataset checks, CTC model bundles, ONNX export and corpus-level PyTorch/ONNX comparison, with explicit input and model-configuration checks.

VSL Streaming Toolkit - Developed a Python library and WebSocket service for ordered keypoints, isolated-sign segmentation, model-input validation and per-connection state.

Vietnamese Lip Reader - Built a Flutter and FastAPI application with video upload, a bounded inference job queue, transcript review and subtitle workflows.

EDUCATION

Ho Chi Minh City Open University Aug 2025-Present

M.Sc. in Computer Science (Artificial Intelligence), in progress; expected 2027.

FPT University, Ho Chi Minh City 2021-2025

Bachelor's degree in Artificial Intelligence.

RESEARCH AND PUBLICATIONS

Focus: visual speech recognition, Vietnamese sign language, multimodal learning and AI for scientific sensing. Six published papers; two first-author papers.

1. **Phuc Nguyen Tran Thien**, Trung Nguyen Quoc, Thai-Binh Tran, Vinh Truong Hoang, Kiet Tran Trung (2026). **Deep Smartphone-Based Classification of NH4+ and NH2- Colorimetric Strips: Revisiting the Role of Explicit Green-Border Calibration**.
IEEE JCSSE 2026 · First author. doi:10.1109/jcsse68839.2026.11597064

2. **Phuc Nguyen Tran Thien**, Trung Nguyen Quoc, Thanh-Danh Nguyen, Vinh Truong Hoang, Kiet Tran-Trung (2026). **Deep NH4+ Regression for Smartphone Colorimetric Test Strips: On the Need for Explicit Green-Border Normalization**.
IEEE JCSSE 2026 · First author. doi:10.1109/jcsse68839.2026.11597039

3. Trung Nguyen Quoc, **Phuc Nguyen Tran Thien**, Vinh Truong Hoang, Kiet Tran-Trung (2026). **Adapting Hierarchical Feature Aggregation and Spatio-Temporal Graph Neural Networks for Vietnamese Visual Speech Recognition on a Small Phrase-Level Dataset**.
IEEE CICN 2026. doi:10.1109/cicn70047.2026.11594397

4. Trung Nguyen Quoc, **Phuc Nguyen Tran Thien**, Vinh Truong Hoang, Kiet Tran-Trung (2026). **Toward Edge-Deployable Vietnamese Lip Reading: A Compact Lipnet-Style Pytorch Recognizer with a Jetson-Oriented Inference Pipeline**.
IEEE CICN 2026. doi:10.1109/cicn70047.2026.11594266

5. Trung Nguyen Quoc, Minh-Vuong Phan, Khoi Pham Dang, Huy Ngo Huu Gia, **Phuc Nguyen Tran Thien**, et al. (2026). **Smartphone-based colorimetric sensing with reference calibration and ensemble machine learning for enhanced detection of nitrite and ammonium ions**.
Analytica Chimica Acta, 1388, 345097. doi:10.1016/j.aca.2026.345097

6. Trương Duy Việt, Ngô Hữu Gia Huy, Phạm Đăng Khôi, **Nguyễn Trần Thiên Phúc** (2025). **MoViNet-A2 cho bài toán nhận diện ký hiệu Tiếng Việt**.
HCMCOU Journal of Science - Engineering and Technology, 20(2), 3–14.
doi:10.46223/hcmcoujs.tech.vi.20.2.4201.2025

CURRENT RESEARCH MANUSCRIPTS

Visual context and text scope in visual speech recognition - research manuscript, public code and numerical reproduction.
Vietnamese isolated-sign recognition and sign-lip fusion - working manuscript, implementation and result archives.

HONORS & AWARDS

1st Prize, AI Track — Cardano SEA Hackathon 2026 · Signova · May 2026.
Top 20 National Finalist - Next-gen Youth Entrepreneurship Challenge 2026 (Vietnam) · Signova · Aug 2026.
Third Prize — Food Innovation and Development 2025 · Thịt Tươi Rói.
Food Security & Social Impact category · HUIT · Dec 2025.
Additional recognition: Top 10 Sparki Innovation Challenge 2022 (Smable); Top 5 Finalist, Youpreneur Launchpad 2024; Top 10, I-Startup 2024.